Exercise Sheet 4

Discriminative Models: Linear and Logistic Regression, Calibrated Scores

Probabilistic Machine Learning

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4.A Least Squares Is a Likelihood

Exercise 4.A.1: Four Flats and a Villa

Model $y = w_0 + w_1 x + \varepsilon, \varepsilon \sim \mathcal{N}(0, \sigma^2)$ i.i.d.; x in m², y in k€.

x	40	60	80	100
y	150	200	280	330

1. Write down $\ell(w_0, w_1, \sigma^2)$ and show that for **every** σ^2 the maximum over (w_0, w_1) is the least-squares fit.
2. Compute $\hat{w}_1 = S_{xy}/S_{xx}$ and $\hat{w}_0 = \bar{y} - \hat{w}_1 \bar{x}$, the residuals and the RSS.
3. Compute $\hat{\sigma}^2_{\mathrm{ML}} = \mathrm{RSS}/n$ and the unbiased $\mathrm{RSS}/(n-2)$. By which factor does the ML version underestimate σ^2 on average?
4. Give the plug-in 95% band $\hat{y} \pm 1.96\hat{\sigma}_{\mathrm{ML}}$ at 80 m² and at 600 m². Then use the half-width that counts the uncertainty about $\mathbf{w}$, $1.96\hat{\sigma}\sqrt{1 + \frac{1}{n} + \frac{(x-\bar{x})^2}{S_{xx}}}$. What do you conclude?

4.B Classification: Only p(y | x)

Exercise 4.B.1: Cross-Entropy and Its Gradient

$h_i = \sigma(\mathbf{w}^\top \mathbf{x}_i)$ with $\sigma(z) = 1/(1+e^{-z})$ and $\ell(\mathbf{w}) = \sum_i [y_i \ln h_i + (1-y_i)\ln(1-h_i)]$.

1. Show $\sigma'(z) = \sigma(z)(1-\sigma(z))$.
2. Show $\frac{\partial \ell}{\partial \mathbf{w}} = \sum_i (y_i - h_i)\mathbf{x}_i$.
3. Six transactions, $\mathbf{x} = (1, \text{new device})$: new device — fraud, fraud, legitimate; known device — fraud, legitimate, legitimate. Compute the gradient at $\mathbf{w} = 0$.
4. Set both components of the gradient to zero: which scores does the MLE give to new and to known devices? Give $\hat{w}_0$, $\hat{w}_1$ and read $\hat{w}_1$ as an odds ratio.

Exercise 4.B.2: The Peak That Runs Away (Python)

```
import numpy as np
x = np.array([-2.0, -1.5, -1.1, -0.6, -0.3, 0.2, 0.6, 1.0, 1.4, 1.9])
y = np.array([0, 0, 0, 0, 0, 0, 1, 1, 1, 1])
sigma = lambda z: 1 / (1 + np.exp(-z))

def fit(tau2=None, steps=10_000, eta=0.05):
    w0 = w1 = 0.0
    for t in range(1, steps + 1):
        h = sigma(w0 + w1 * x)
        g0 = ___                        # gradient for the intercept
        g1 = ___                        # gradient for the slope
        if tau2 is not None:
            g1 -= ___                   # log-prior N(0, tau2) on the slope
        w0, w1 = w0 + eta * g0, w1 + eta * g1
        if t in (10, 100, 1_000, 10_000):
            h = sigma(w0 + w1 * x)
            print(t, round(w1, 2), np.sum(y * np.log(h) + (1 - y) * np.log(1 - h)))
```

1. Fill in the blanks and run `fit()`. What happens to the slope, the log-likelihood and the scores? Why will it never stop?
2. Run `fit(tau2=1)` and `fit(tau2=100, steps=100_000)`. Explain the difference with the MAP formula of the lecture.

4.C Generative vs Discriminative

Exercise 4.C.1: The Same Line

Naive Bayes with yes/no flags, $\theta_{jy} = P(x_j = 1 \mid y)$, prior $\pi = P(y = 1)$. Lecture numbers: $\pi = 1/1000$; new device 0.6 vs 0.1; foreign IP 0.7 vs 0.05; night 0.5 vs 0.2 (fraud vs honest).

1. Show that $\log \frac{P(y=1|x)}{P(y=0|x)} = w_0 + \sum_j w_j x_j$ and give w_j and w_0.
2. Compute the three weights and w_0, and check that a transaction with all three flags gets $P(\text{fraud} \mid x) \approx 17.4\%$.
3. The foreign-IP flag is **missing**. Compare what naive Bayes should do with plugging $x_{\text{IP}} = 0$ into the linear form.
4. Why can logistic regression, which has the same linear form, not do what naive Bayes does in 3?

Exercise 4.C.2: Counting Twice, in Code

Complete the three marked lines (numpy only), then run the script.

```
import numpy as np
rng = np.random.default_rng(42)
n = 5000
y = (rng.random(n) < 0.3).astype(int)                        # 1 = fraud
flag = lambda pf, ph: (rng.random(n) < np.where(y == 1, pf, ph)).astype(int)
device = flag(0.6, 0.1)
copy = np.where(rng.random(n) < 0.95, device, 1 - device)   # a near-copy
night = flag(0.5, 0.2)
X = np.column_stack([device, copy, night])

def fit_nb(X, y, alpha=1.0):
    prior = ...                     # (a) P(y = 1)
    th1 = ...                       # (b) Laplace-smoothed P(x_j = 1 | y = 1)
    th0 = ...                       #     and P(x_j = 1 | y = 0)
    w = np.log(th1 / th0) - np.log((1 - th1) / (1 - th0))
    w0 = np.log(prior / (1 - prior)) + np.log((1 - th1) / (1 - th0)).sum()
    return w0, w

def fit_lr(X, y, steps=5000, lr=0.5):   # gradient ascent, as in the lecture
    Xb = np.column_stack([np.ones(len(X)), X]); w = np.zeros(Xb.shape[1])
    for _ in range(steps):
```

1. Print w_0, w and $P(\text{fraud} \mid x = (1, 1, 1))$ for both models.
2. Compare the weights of `device` and `copy`. Which model counts the device twice, and which shares the credit?
3. Which probability is closer to the fraction of frauds among the training rows with $x = (1, 1, 1)$?

4.D When Is a Score a Probability?

Exercise 4.D.1: The Rebalanced Fraud Model

A model is trained on a 50/50 sample; on real traffic fraud is **1 in 1,000**. Undersampling kept every fraud and a random share of legitimate cases.

1. Use Bayes' theorem in odds form to show: real odds = model odds × $\frac{1/999}{1}$. Which assumption about $p(\mathbf{x} \mid y)$ do you need?
2. Convert the scores 0.5, 0.9 and 0.99 into real fraud probabilities.
3. Blocking a legitimate card costs **€10**, missing a fraud **€500**. Above which real probability should a card be blocked, and which **raw score** is that?

Exercise 4.D.2: Weight Decay Is a Prior

1. With $y_i = \mathbf{w}^\top \mathbf{x}_i + \varepsilon_i, \varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ and a prior $\mathbf{w} \sim \mathcal{N}(0, \tau^2 \mathbf{I})$, show that the MAP minimises $\mathrm{RSS}(\mathbf{w}) + \lambda \|\mathbf{w}\|^2$ with $\lambda = \sigma^2/\tau^2$.
2. One weight, no intercept, $y = wx + \varepsilon$. Show $\hat{w}_{\mathrm{MAP}} = \dfrac{\sum_i x_i y_i}{\sum_i x_i^2 + \lambda}$.
3. Data $(x, y) = (1, 2), (3, 7)$, $\sigma^2 = 1, \tau^2 = 0.1$. Compute the MLE and the MAP. What happens as $\tau^2 \to \infty$ and as $\tau^2 \to 0$?